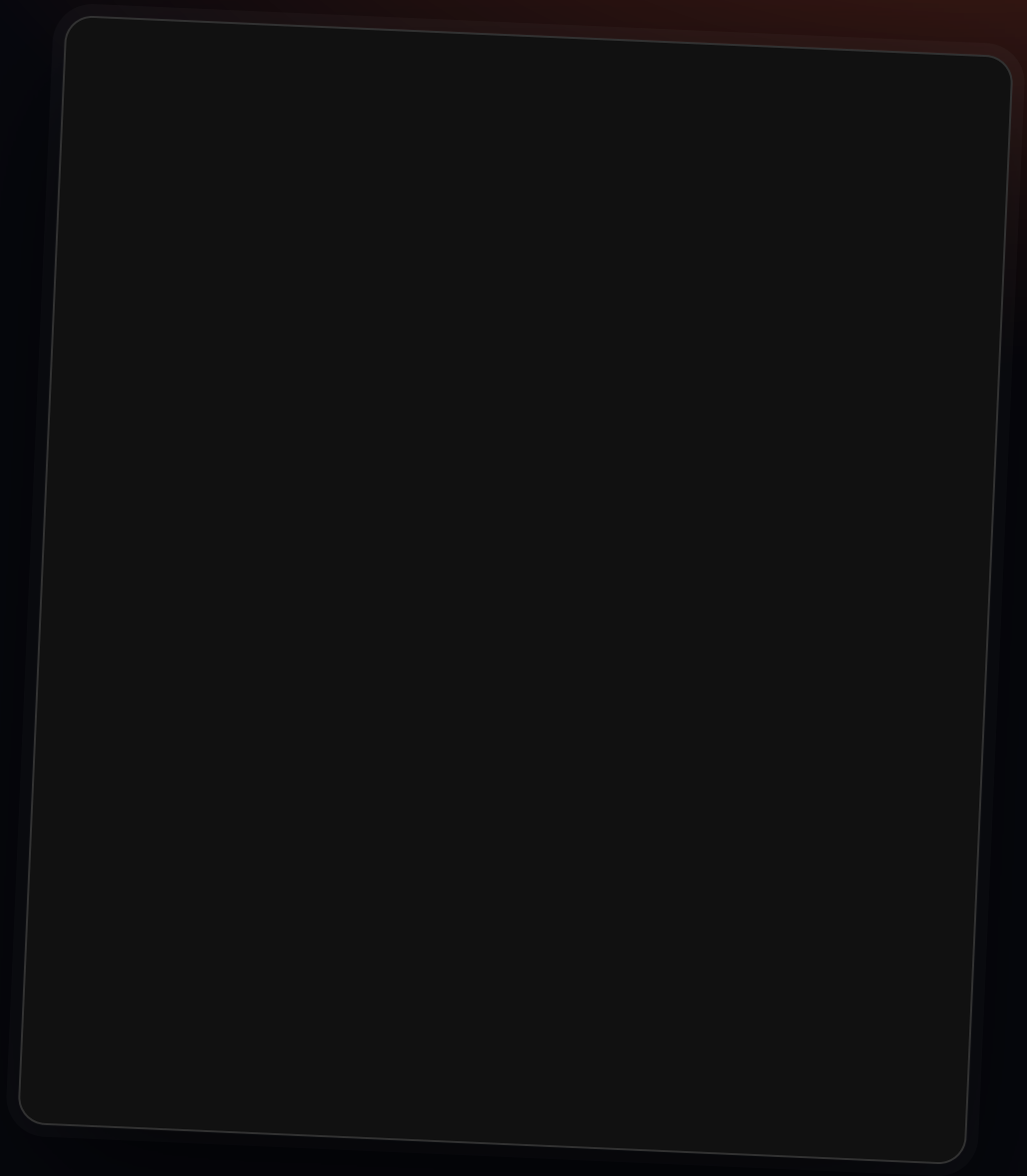
SWISSQUOTE WALLET TRACK · FERRATA

Clients who want their own keys leave the bank.

Today they withdraw to an external wallet and leave the ecosystem. Ferrata gives them their keys, one step from their account.

- Passkey, no seed phrase
- Bank-grade risk analysis
- Recovery without the bank alone



CUSTODY MODEL · CREATION · AUTHENTICATION

2-of-2 where only you can spend

Your passkey

Created on your device, Face ID / fingerprint. Signs every payment. REAL

Your recovery phone

Paired by QR, its key registered on the account. REAL

The bank

Analyses risk, signs a receipt, attests KYC. Holds no key to your funds.

Create

Log in to the bank account → passkey on the device → it signs a bank challenge → wallet bound to the client ID

WHO, ALONE…	SEND	BLOCK	CHANGE KEY
Your passkey	✓	✗	✗
Recovery phone	✗	✗	✗
The bank	✗	✗	✗
Phone + bank KYC + delay	✗	✗	✓ you can cancel

Low	Medium	High	Sanctions
one touch	2nd confirmation	explained score + "I understand"	strongest warning, decision recorded

Pros

No seed phrase · the bank can neither move nor freeze funds · works if the bank is down

Cons

Lost passkey + phone → needs the bank's KYC · risk checks are advice, not a lock

THREAT MODEL · RECOVERY

Lost passkey? Phone + identity + time.

1. **Pair a phone**
   QR + fingerprints checked on both screens REAL
2. **"I lost my passkey"**
   new passkey on the new device, QR to the phone REAL
3. **Phone signs + KYC**
   ID document MRZ, age, expiry REAL · video, face, NFC SIMULATED
4. **Smart delay**
   base ×1 to ×14 from account history, reasons on both devices REAL
5. **Rotation**
   needs phone signature *and* bank attestation, on chain REAL

Phishing site

Explained risk score and warning before signing

Stolen phone + email

Phone alone can't send. Recovery still needs KYC; the old passkey gets an alert and cancels

The bank, alone or with a partner

Cannot move, freeze or rotate: the phone's signature is required on chain

Bank gone for good

The passkey alone keeps moving funds. No bank signature needed

Known limit: if the attacker retries after a cancel ("contested"), the delay grows but the old passkey can no longer cancel; KYC is then the last barrier, and it is simulated today.

ARCHITECTURE · REAL VS MOCKED

Real chains, every transfer traceable

CLIENT

Chrome extension
passkey, sends, dApp provider

Phone app
recovery key, KYC

TESTNETS · SIGNED AND BROADCAST

Base Sepolia · gasless USDC
ERC-4337 smart account, Pimlico paymaster pays the gas

Solana devnet
Ed25519

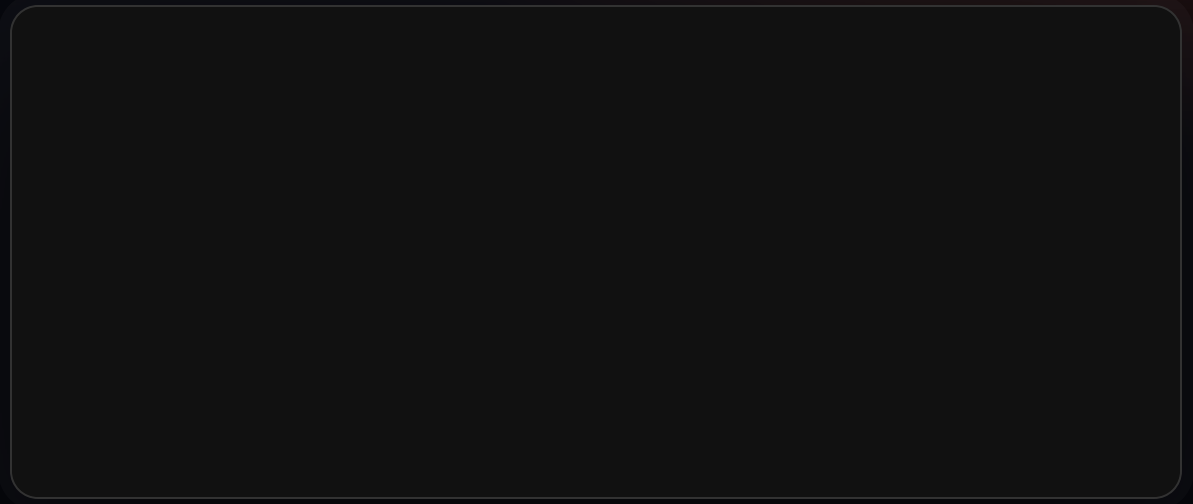
BANK SERVICES · RUST

Risk engine · receipts · audit log · recovery attestation
connectors read Bitcoin, Ethereum, Solana, TON

Compliance provider
KYC, sanctions list, exposure graph MOCKED

"Ferra" ledger
account rules, signed blocks, replay-verified; mirrors every network's activity OWN CHAIN

COMPLIANCE DESK · FERRA VISUALIZER



Replays every block
Scores each transfer
Risk by network
Sanctioned & reported addresses
Follows the money
Exports for auditors

LEGAL · BUSINESS

Software, not custody: the bank can never act alone

Custody · Art. 242a DEBA

Only the client's passkey can spend; the bank cannot move, freeze or rotate the funds alone, nor with a partner. No power of disposal, so not custody.

To be validated with compliance.

Travel Rule · FINMA 02/2019

Ownership proved by construction: the passkey signs a bank challenge at creation, bound to the identified client. Every transfer is recorded with sender, recipient, asset and amount.

Sending institution for third-party payments? To be validated.

Sanctions

The bank warns and records the decision, but does not refuse, because it does not hold the key.

Whether that is enough under Swiss sanctions law: to be validated with compliance.

What the bank gains

Keeps the crypto client and the flow back to the account · paid analysis tier · gasless transfers billed in CHF · fees on compliant DeFi · KYC recovery nobody else offers · no client keys to insure

723 M	85.7 M	−12.1 %
CHF revenue 2025	CHF from crypto (≈12 %)	crypto volume 2025

Reported FY2025 figures (press coverage of Swissquote results). What share of crypto clients would self-custody is an assumption to validate.

BONUS · LIMITS · CONCLUSION

Gasless · done

USDC on Base Sepolia through an ERC-4337 account; Pimlico's paymaster pays the gas under a sponsorship policy. The client never holds ETH.

dApps · partial

EIP-1193 / 6963 provider, origin from the browser; the bank's site reads the wallet and sends through the Ferrata window. Third-party dApp signing waits for the approval window.

Simulated

Video KYC, face match, NFC chip. The "Ferra" ledger enforces the account rules; the testnets carry real transfers.

Your key. Your money.
A bank that has your back,
not your keys.